

Cricket Player Detection and Tracking System

1. Introduction

This project implements a cricket player detection and tracking system. It detects players from video frames and assigns persistent tracking IDs using modern deep learning and tracking algorithms.

In addition to core detection and tracking, the system includes advanced analytical features such as player trajectory visualization, heatmap generation, and homography-based top-view transformation. These optional modules provide deeper insights into player movement patterns and spatial positioning, enhancing the overall analytical capabilities of the system.

3. Model Architecture

The detection backbone is based on the **YOLO12x** architecture. This model was chosen because of its strong performance in detecting small and fast-moving objects, which is crucial for capturing cricket players in broadcast footage.

The model processes each video frame and outputs bounding boxes along with unique IDs for detected players.

4. Tracking Methodology

To assign and maintain consistent player identities, the system uses **BoT-SORT** (Better Online Tracking with SORT). The tracker integrates multiple techniques:

- **Kalman Filters** for motion prediction
- **IoU-based matching** for spatial consistency
- **Appearance-based feature embeddings (ReID)** for identity preservation

This ensures that each player retains their tracking ID even after occlusions or overlaps.

5. GPU Optimization

The system is optimized to run on GPU using CUDA . GPU acceleration significantly improves inference speed, allowing real-time or near-real-time processing of high-resolution cricket videos.

6. Additional Features

Beyond basic detection and tracking, the system also implements:

- **Player trajectory visualization** to show movement paths
- **Homography-based transformations** for better spatial understanding
- **Advanced visual overlays** for enhanced interpretability
- **Top-View(bird's-eye) projection**

These features improve the analytical value of the system.

7. Challenges Faced and Solutions

Challenge	Solution Implemented
Player occlusion	Track buffering and ReID embeddings
Fast motion	Kalman filter-based prediction
Spectators	Applied Masking

8. Results

The system successfully detects and tracks multiple players simultaneously while maintaining consistent IDs throughout the video. The output video clearly shows bounding boxes and tracking IDs for each player with smooth and stable tracking performance.

9. Future Improvements

Future enhancements can include:

- Jersey number recognition
- Team-wise clustering

10. Conclusion

The project demonstrates an efficient and robust solution for cricket player detection and tracking using deep learning-based object detection and modern multi-object tracking techniques. The system is suitable for real-time sports analytics and lays a strong foundation for advanced cricket analysis tools.